

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Data Engineering professional with **5+ years of experience** designing, developing, and maintaining scalable enterprise data solutions using Python, SQL, PySpark, Apache Spark, Databricks, Delta Lake, and modern cloud-based data platforms.
- Hands-on experience building and supporting **ETL/ELT pipelines, Delta Lake tables, and Medallion Architecture layers (Bronze, Silver, Gold)** for reliable ingestion, transformation, curation, and consumption of enterprise data.
- Experienced ingesting data from **enterprise applications, REST APIs, relational databases, flat files, and cloud storage**, transforming raw data into validated and analytics-ready datasets using PySpark and SQL.
- Strong expertise in **data transformation, validation, cleansing, reconciliation, data quality, data modeling, and pipeline monitoring**, ensuring accuracy, consistency, reliability, and usability of production datasets.
- Supported **reporting, analytics, business intelligence, AI, and ML initiatives** by developing curated data products and scalable data-processing workflows aligned with business and analytical requirements.
- Experienced implementing **data governance, security, documentation, Git version control, CI/CD, monitoring, troubleshooting, and cloud data engineering best practices** across collaborative Agile environments.
- Strong analytical and problem-solving professional with experience collaborating with **Data Engineers, Data Architects, Analytics Engineers, product teams, and business stakeholders** while demonstrating initiative, accountability, adaptability, customer focus, and continuous learning.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Data Processing:** Python, SQL, PySpark, Apache Spark, Spark SQL, Java, Pandas, DataFrames, Distributed Processing, Batch Processing, Data Transformation.
- **Databricks & Lakehouse:** Azure Databricks, Databricks Lakehouse Platform, Delta Lake, Delta Tables, Medallion Architecture, Bronze/Silver/Gold Layers, Databricks Workflows, Lakehouse Architecture.
- **Data Engineering:** ETL/ELT, Data Ingestion, Data Transformation, Data Cleansing, Data Validation, Data Quality, Data Modeling, Data Integration, Data Reconciliation, Structured/Semi-Structured Data.
- **Sources & Storage:** REST APIs, Enterprise Applications, Relational Databases, SQL Server, PostgreSQL, MySQL, Oracle, Cloud Storage, JSON, CSV, Parquet, Delta Format.
- **Governance & Analytics:** Data Governance, Data Security, Data Quality Standards, Metadata, Business Intelligence Concepts, Reporting, Analytics, Curated Data Products, AI/ML Data Enablement.
- **DevOps & Operations:** Git, Azure DevOps, CI/CD, Jenkins, Docker, Cloud Platforms, Pipeline Monitoring, Logging, Troubleshooting, Performance Optimization, Production Support, Technical Documentation.
- **Delivery & Collaboration:** Agile/Scrum, Sprint Planning, Sprint Reviews, Problem Solving, Critical Thinking, Team Collaboration, Written/Verbal Communication, Time Management, Attention to Detail, Customer Focus, Continuous Learning.

EXPERIENCE

Software Engineer AI

Oct 2024 – Present

PCMI LLC (Full Time)

Dallas, TX, USA

- Designed and maintained enterprise **Databricks Lakehouse data solutions** using Python, PySpark, Apache Spark, SQL, Delta Lake, and cloud services, supporting insurance reporting, analytics, operational decision-making, and AI-driven data use cases.
- Built scalable **ETL/ELT pipelines and Medallion Architecture layers** across Bronze, Silver, and Gold datasets, ingesting data from enterprise applications, REST APIs, relational databases, files, and cloud storage into curated analytical data products.
- Developed reusable **PySpark and SQL transformations** for cleansing, standardization, enrichment, aggregation, deduplication, and reconciliation while using Delta tables to maintain reliable and performant production datasets.
- Implemented comprehensive **data validation, data quality, and governance controls** covering completeness, accuracy, consistency, duplicate handling, schema validation, source-to-target reconciliation, and secure access to sensitive enterprise data.
- Monitored production pipelines, analyzed Spark execution behavior, and resolved **processing failures, performance bottlenecks, data inconsistencies, and dependency issues**, improving reliability and operational support of data workflows.
- Collaborated with **Data Architects, Analytics Engineers, product owners, source application teams, and business stakeholders** to translate reporting and analytics requirements into scalable data models, transformations, and consumption-ready datasets.
- Applied engineering best practices using **Git, Azure DevOps, CI/CD, documentation, Agile sprint planning, code reviews, logging, and reusable framework components**, while continuously improving data-platform reliability and maintainability.

Environments: Python, SQL, PySpark, Apache Spark, Azure Databricks, Databricks Lakehouse, Delta Lake, Medallion Architecture, Bronze/Silver/Gold, ETL/ELT, REST APIs, Cloud Storage, Data Modeling, Data Validation, Data Cleansing, Data Quality, Data Governance, Git, Azure DevOps, CI/CD, Monitoring, Troubleshooting.

Software Engineer AI

Dec 2023 – Sept 2024

Client: CVS

Remote, USA

- Designed and developed cloud-based **healthcare data solutions** using Python, PySpark, Apache Spark, SQL, and Databricks concepts, supporting analytics, reporting, and enterprise healthcare data-processing requirements.
- Built **ETL/ELT ingestion pipelines** to integrate structured healthcare information from REST APIs, relational databases, files, and cloud repositories, transforming raw source data into standardized and curated datasets.
- Developed PySpark and SQL workflows for **data transformation, cleansing, validation, enrichment, deduplication, and aggregation**, improving consistency and usability of high-volume healthcare data consumed by analytical applications.
- Applied **data modeling and Delta Lake concepts** to organize datasets into raw, refined, and curated layers, supporting scalable consumption patterns for business intelligence, analytics, and downstream data products.
- Implemented **data quality, governance, security, and reconciliation controls** to identify missing records, inconsistent values, schema issues, duplicate data, and source-to-target discrepancies across integrated data flows.
- Monitored data workflows and investigated **pipeline failures, processing delays, SQL performance issues, dependency problems, and data inconsistencies**, documenting findings and implementing reliable corrective solutions.
- Collaborated with architects, data engineers, analytics teams, and business stakeholders through **Agile planning, requirements analysis, sprint reviews, technical documentation, Git-based development, CI/CD, and production delivery**.

Environments: Python, SQL, PySpark, Apache Spark, Databricks Concepts, Delta Lake, ETL/ELT, REST APIs, PostgreSQL, MySQL, Cloud Storage, Data Modeling, Data Transformation, Data Validation, Data Cleansing, Data Quality, Git, CI/CD, Monitoring, Reporting, Analytics, Agile/Scrum.

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Developed enterprise **data engineering and analytics solutions** using Python, PySpark, Apache Spark, SQL, and relational databases, supporting large-scale operational applications and business intelligence workloads.

- Built reusable **ETL/ELT pipelines** to ingest data from databases, APIs, enterprise applications, and files, transforming raw source information into standardized datasets for reporting and analytical consumption.
- Implemented PySpark and SQL transformations for **data cleansing, validation, aggregation, enrichment, normalization, and business-rule processing**, improving the accuracy and consistency of downstream enterprise datasets.
- Designed relational and analytical **data models** using PostgreSQL, MySQL, and Oracle, organizing business entities and relationships into structures optimized for reporting, analytics, and application consumption.
- Implemented practical **data quality and governance standards** including schema validation, completeness checks, duplicate identification, reconciliation, exception management, and technical documentation for recurring data workflows.
- Monitored batch-processing pipelines and resolved **data issues, transformation failures, SQL performance bottlenecks, and integration problems**, applying root-cause analysis and performance optimization techniques.
- Worked with engineers, architects, analysts, and product teams using **Git, Jenkins, CI/CD, Agile planning, sprint reviews, documentation, and collaborative engineering practices** to deliver scalable and reliable data solutions.

Environments: Python, SQL, PySpark, Apache Spark, PostgreSQL, MySQL, Oracle, ETL/ELT, Data Ingestion, Data Transformation, Data Modeling, Data Validation, Data Cleansing, Data Quality, REST APIs, Git, Jenkins, CI/CD, Monitoring, Documentation, Reporting, Analytics.

Associate Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed enterprise **data-processing and integration solutions** using Python, SQL, relational databases, and backend technologies, supporting operational reporting and analytics across distributed client environments.
- Built reusable **data ingestion and ETL workflows** to extract information from enterprise applications, relational databases, REST APIs, and files and transform raw records into structured downstream datasets.
- Developed Python and SQL components for **data transformation, cleansing, validation, preprocessing, aggregation, and reconciliation**, improving accuracy and consistency across recurring business data processes.
- Supported **data modeling and relational schema design** using PostgreSQL and MySQL, creating structured datasets and reusable database objects for reporting, operational applications, and analytics consumption.
- Applied **data quality and governance practices** including completeness checks, duplicate detection, exception handling, schema validation, access considerations, and documentation of critical data workflows.
- Investigated **data inconsistencies, SQL issues, processing failures, and integration problems**, using analytical problem solving, logging, monitoring, and root-cause analysis to restore reliable data processing.
- Collaborated with engineers, architects, analysts, and stakeholders through **Git-based development, CI/CD, Agile ceremonies, technical documentation, teamwork, knowledge sharing, and continuous improvement** of enterprise data solutions.

Environments: Python, SQL, PostgreSQL, MySQL, ETL, Data Ingestion, Data Transformation, Data Cleansing, Data Validation, Data Quality, Data Modeling, REST APIs, Enterprise Applications, Git, CI/CD, Logging, Monitoring, Reporting, Analytics, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.